GANAPATHI RAMKUMAR PALANIVEL

AI Architect | Generative AI Architect | Solution Architect | Cloud & Enterprise Architecture

+91-7996656111 | pganapathiramkumar@gmail.com | https://www.linkedin.com/in/palaniram/ | www.ramvector.com

GenAI Systems | RAG & Multi-LLM Architecture | Engineering & Manufacturing AI | 1000+ Users | 60% Faster Delivery

PROFILE SUMMARY

Enterprise AI Architect and **Solution Architect** with **14+ years** designing and delivering scalable **AI platforms**, **multi-cloud architectures**, and **enterprise transformation programs**. Currently **AI Architect at Creative Synergies Group**, leading **AI-driven engineering automation** across CAD/PLM platforms (Autodesk Inventor, Vault, Siemens NX) with the President and senior leadership. Architected production **GenAI systems**, driving **60% delivery acceleration** across **1000+ enterprise users**. Built and deployed **RamVector**, a full-stack **RAG AI assistant platform** - demonstrating **production-grade AI architecture**. Deep expertise in end-to-end **AI solution design**; **TOGAF / Zachman aligned**, **AZ-305 certified**, **Anthropic-certified**, with **cross-functional team leadership** (**15 engineers**) across **UK** and **India**.

- **AI Platform & GenAI Architecture:** Designed end-to-end GenAI reference architectures including RAG pipelines, vector store selection (Pinecone / pgvector), prompt orchestration, guardrails, observability, and human-in-the-loop evaluation across enterprise workloads.
- **Responsible AI Governance & Compliance:** Defined AI governance frameworks covering model risk management, bias evaluation, data lineage, and compliance for enterprise GenAI rollouts.

FEATURED AI PROJECT

Vision React Code Generation System - AI Architecture & Product Innovation

- Architected a multi-modal GenAI platform converting UI images and PDF mockups into production-ready React / Angular code - reducing front-end development effort from ~2 weeks to 5–10 minutes (60–70% efficiency gain) and lowering delivery cost by ~40% through automation of manual front-end effort.
- Defined the end-to-end solution architecture: multi-modal input pipeline, LLM-based component mapping engine, code generation module, structured output parsing, and validation layer.
- Submitted for peer-reviewed publication: Zenodo and ORCID.

CORE COMPETENCIES

AI & GenAI Solution Architecture · Artificial Intelligence · Machine Learning · Deep Learning · NLP · LLM Systems Design · RAG Pipeline Architecture · Vector Database Integration · Agentic AI & Multi-Agent Systems · MLOps & AI Platform Engineering · AI Governance & Responsible AI · Retrieval Evaluation & Hallucination Mitigation · Prompt Engineering at Scale · Model Context Protocol · CAD / PLM AI Automation · Engineering Rules Automation · Engineering Data Traceability · CAD ↔ BOM Reconciliation · Enterprise Architecture · Architecture Review & Governance · Strategic Planning · Technology Roadmap · Digital Transformation · Business Alignment · C-Suite Communication · Stakeholder Governance · Time-to-Market · Cost Optimization · Greenfield Architecture · Multi-Cloud Design · HLD / LLD · Microservices, API Strategy & Integration Design · DevOps, IaC & CI/CD Transformation · Cross-Functional Leadership & Delivery · Solutioning & Pre-Sales Architecture Support

TECHNICAL SKILLS

AI & GenAI Platforms: GPT-4, Claude Sonnet / Haiku (AWS Bedrock), Azure OpenAI Service, Azure AI Foundry, Azure AI Document Intelligence, Azure AI Search, AWS SageMaker

LLM Engineering & Architecture: RAG Architecture, Fine-Tuning vs RAG Design, Prompt Engineering, Token Optimisation, Model Selection, Embedding Models, Hallucination Mitigation, Guardrails, Agentic AI, Multi-Agent Patterns, Model Context Protocol (MCP), Function Calling, Tool Use

CAD / PLM & Engineering AI: Autodesk Inventor, Vault, Apprentice Server (Python & APIs), Siemens UG NX, Siemens Tecnomatix Process Simulate, FBOM, PLM / ERP / MES Integration, Macro-to-AI Automation, ML / Deep Learning / NLP, ML-derived Deterministic Rules

Engineering Data & Knowledge: Knowledge Graphs, GraphRAG, GD&T-aware Extraction, Computer Vision / OCR for Drawings, Change-Impact Analysis

AI Frameworks & Orchestration: LangChain, LangGraph, LlamaIndex, Semantic Kernel, AutoGen, CrewAI

Vector Databases & Retrieval: Pinecone, pgvector, Chroma, Retrieval Evaluation

Evaluation, Monitoring & Observability: Ragas, DeepEval, LangSmith, Arize, Weights & Biases, AI Observability, Model Monitoring

Responsible AI & Governance: Responsible AI, AI Governance Frameworks, NIST AI RMF, ISO/IEC 42001, Compliance, Risk Controls

Cloud Platforms: Azure (VMs, App Services, Functions, SQL, Logic Apps, API Management, Key Vault, ARM), AWS (Bedrock, SageMaker, EC2, Lambda)

Architecture & Design: HLD / LLD, Enterprise Architecture (TOGAF / Zachman aligned), Event-Driven Architecture, API Strategy, PoC-to-Production Delivery, Greenfield Architecture, Cost Optimization

DevOps & Infrastructure: GitHub Actions, Azure DevOps, Jenkins, Docker, Kubernetes, PowerShell

Development Technologies: .NET Core 8, ASP.NET MVC, React.js, Node.js, Python, C#, REST APIs

Power Platform: Power Automate, Power Apps, Copilot Studio

Security & Access Management: IAM, Zero Trust, Azure Key Vault

WORK EXPERIENCE

Sep'26 – Till Date: **AI Architect** | **Creative Synergies Group, Bangalore**

Key Result Areas:

- Spearhead AI transformation with the **President, VP, AVP, GM and AGM** across **5 business units**, guiding **200+ engineers** from manual effort to automation through a phased **AI roadmap (Discover → Prioritise → MVP → Scale)**.
- Reduced engineering effort from **~1 day to ~5 minutes** by moving manual work to automation and wider **AI solutions**, converting legacy **VB / C# macros** (including Siemens UG NX).

- Architect AI automation across **Autodesk Inventor, Vault and Apprentice Server** using **Python and APIs** - component similarity, **duplicate-part detection** and **BOM comparison**.
- Designed a **CAD → Drawing → FBOM traceability pipeline** capturing part hierarchy, critical dimensions, tolerances, **GD&T** and material, with **engineering-criticality scoring (P0–P3)**, and a **Graph + RAG knowledge model** for **change-impact analysis and CAD ↔ BOM reconciliation**.
- Automate **welding, hole fastening and interference validation** (supported by **Process Simulate**) using hybrid **RAG, ML-derived deterministic rules, deep learning and NLP** under an **"AI recommends, rules validate, humans approve"** model.
- Assessed feasibility of **Azure AI Foundry, Document Intelligence and AI Search** for engineering document extraction and retrieval, and authored the **high-level design for the proposal**.
- Established **AI governance and security** standards (**Zero Trust, NIST AI RMF, ISO/IEC 42001**) to scale AI safely across business units.

Feb'22 – May'26: **Lead Digital Engineer - Solution Architect | Sonata Software, Bangalore (Onsite: Belfast, UK)**

Key Result Areas:

- Architected a **production GenAI platform** (Claude on AWS Bedrock, Azure OpenAI) with a custom spec → component mapping → React code → validation toolchain, **accelerating WinForms-to-React modernisation by 60% for 1000+ enterprise users**.
- Led **pre-sales architecture** for AI-led modernisation deals - solution blueprints, ROI models and effort estimates that **secured enterprise GenAI engagements**.
- Defined **GenAI solution patterns** (RAG, agentic workflows, multi-model orchestration) with cost, latency and accuracy trade-off analysis, including enterprise **RAG** with pgvector and LangChain.
- Established an **AI governance framework** - responsible AI controls, model access policies and audit logging - enabling compliant Azure OpenAI and AWS Bedrock deployment for a **UK financial services client**.
- Delivered **greenfield architecture** for new digital platforms, defining target-state designs, technology stacks and integration blueprints.
- Led a **1,000+ Windows Server migration (2012 → 2019) with zero downtime**, coordinating a **15-engineer team**; received formal client appreciation from Infosys.
- Modernised CI/CD from TeamCity to **GitHub Actions** with embedded DR controls, and defined the target-state blueprint for an **OutSystems** legacy platform.

Nov'20 – Feb'22: **Senior Application Developer | TE Connectivity, Bangalore**

- Engineered enterprise project tracking platform with multi-feed data pipeline processing, reducing reporting latency for global operations teams.

Aug'17 – Nov'20: **Senior Application Developer | Khusheim Industrial Equipment, Bangalore (Onsite: Dammam, Saudi Arabia)**

- Delivered full-cycle ERP implementation (Trio ERP) - requirement gathering, solution design, integration, and production go-live - managing direct C-suite stakeholder engagement across the Saudi Arabia engagement.

May'17 – Aug'17: **Senior Application Developer | Unisys, Bangalore**

- Delivered RPA and chatbot automation solutions using CI/CD pipelines, reducing manual processing effort across enterprise operations.

Apr'16 – Feb'17: **Application Developer | Cognizant (CTS), Chennai**

Mar'15 – Nov'15: **Software Engineer | Volen Software Services, Bangalore**

Jul'14 – Feb'15: **Software Engineer | HSR Industrial Consultants, Bangalore**

Dec'11 – Jun'14: **Analyst | Zenta, Chennai**

AI PROJECTS & INNOVATION

RamVector - Enterprise RAG AI Assistant Platform | Personal Production Project | 2024 – Present

App Store | Website | Live Demo | GitHub

- Built and deployed a production RAG assistant on **iOS and Android** - 7-service microservices architecture (FastAPI, React, pgvector, PostgreSQL, Redis, RabbitMQ) with multi-LLM fallback and full observability.

GenAI Migration Platform - WinForms to React | Sonata Software | 2023 – 2026

- Designed custom AI migration toolchain with Claude (AWS Bedrock) and GPT-4: spec generation → component mapping → React code output → quality validation - achieving 60% faster enterprise UI delivery across 1000+ users.

CERTIFICATIONS

Azure Solutions Architect Expert (AZ-305) - Valid through Jun 2027
DevOps Engineer Expert (AZ-400) - Valid through Jun 2027
Azure Administrator Associate (AZ-104) - Valid through Jun 2027
Azure Network Engineer Associate (AZ-700) - Valid through Jun 2027
Azure Virtual Desktop Specialty (AZ-140) - Valid through Jun 2027
CPMAI - Cognitive Project Management in AI (PMI) (Jul 2025)
Anthropic Certified: Model Context Protocol: Advanced Topics (2026)
Anthropic Certified: Building with Claude API (2026)
Anthropic Certified: Claude Code in Action (2026)
Anthropic Certified: Claude Code 101 (2026)

EDUCATION

M.Tech, Data Science & Machine Learning — PES University | Completed 2026
CTO Leadership Programme — IIM Lucknow | Expected Nov 2026
Doctor of Business Administration — Walsh College, USA | Expected 2028
B.Tech, Information Technology — Anna University | 2009